

CWT-DTAS Series Duct Type Air Velocity Transmitter Manual (RS485 type)



Introduction:

This transmitter is designed to measure the wind speed in the pipeline. The high-precision wind speed measurement unit independently developed by our company is sensitive to reaction and can quickly and accurately measure small wind speeds. It has the characteristics of good stability, small drift and high precision. The transmitter can calculate the real-time air volume value according to the cross-sectional area of the pipeline set for measurement. The pipeline type adopts the installation method, which is convenient for on-site installation. The product adopts anti-interference circuit design, which can withstand various strong electromagnetic interferences such as on-site frequency converters. The equipment adopts imported screw-free terminal wiring, which is more convenient to use.

Features:

- High-precision wind speed measurement unit, low starting wind speed, sensitive response, suitable for ventilation ducts, fume ducts and other harsh environments
- Adopt the full-range secondary calibration method, with good linearity and high accuracy
- Open-hole flange installation, high-quality silicone sealing ring, small air leakage, durable
- The cross-sectional area of the measuring pipe can be set, and the real-time air volume value can be uploaded according to the set cross-sectional area of the pipe
- 485 communication interface, standard Modbus-RTU protocol, communication address and baud rate can be set, and the communication line can be up to 2000 meters
- The device address supports two methods: software configuration and DIP switch setting
- The accuracy of the range 0~10m/s is $\pm(0.1+2\%FS)$; the accuracy of the range 0~15m/s, 0~20m/s, and 0~30m/s is $\pm(0.2+2\%FS)$ m/s
- The wiring terminals use imported screw-free terminals, no tools are required, just press and plug to connect
- Adopting special EMC anti-interference components, it can withstand various strong electromagnetic interferences such as on-site inverters
- With wind speed direction marking, it is easy to install on site

Specification:

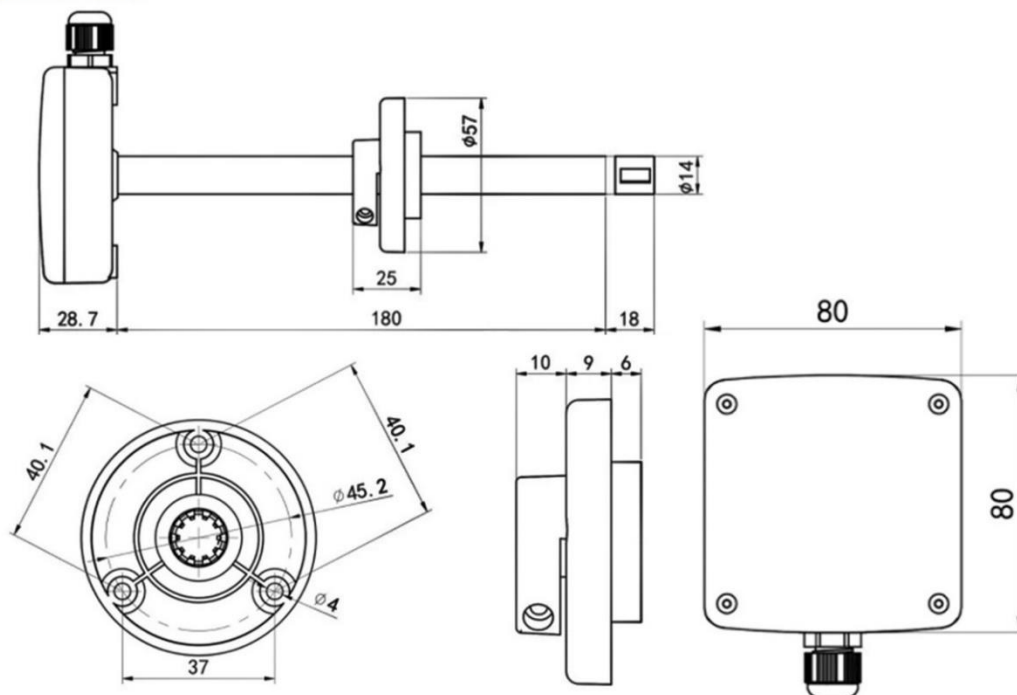
power supply	10-30V DC
Max power consumption	0.6W
Measurement medium	Air, nitrogen, oil fume exhaust gas
Accuracy	$\pm (0.2+2\%FS)$ m/s
Operating temperature	-40℃~+60℃
Output	4~20mA
Resolution	0.1m/s
Response time	2s
Long-term stability	$\leq 0.1\text{m/s/y}$

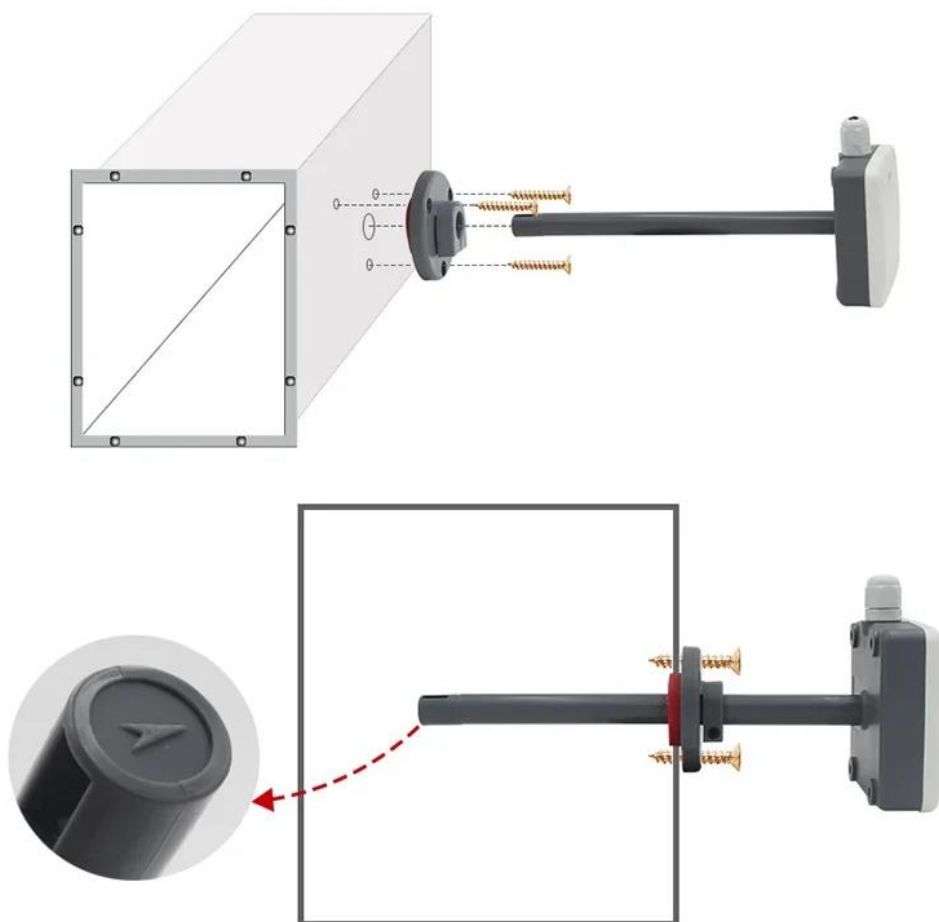
Wiring:



NO.	Internal identification	Description
1	A	RS485 A+
2	V+	Power+
3	GND	Power-
4	B	RS485 B-

Product size:





RS485 communication

Default parameters: 4800,n,8,1

Default device address is 1

Modbus RTU protocol

Read status registers, read function code: 0x03					
Register address (Hex)	PLC Address (decimal)	meaning	Number of bytes	content	remark
0000	40001	Air velocity	2	0.1 m/s	read
0001	40002	Air volume	2	0.1 m ³ /h	read
0002	40003	Air volume high 16 byte	2	1 m/s	Read (Float 32)
0003	40004	Air volume low 16 byte	2		

Parameters registers, read function code: 0x03, write function code: 0x06					
0200	40513	Pipe cross-sectional area	2	1 cm ²	0~65535
07D0	42001	Slave ID	2		1-254
07D1	42002	baud rate	2		0: 2400 1: 4800 (default) 2: 9600 3: 19200 4: 38400 5: 57600 6: 115200 7: 1200

E.g., master read Air velocity:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x00	0x00	0x00	0x01	0x84	0X0A

Sensor responds:

Address	Function Code	Number of byte	Velocity value	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x02	0x00 0x6C	0xB8	0x69

Velocity calculates:

006C H= 108 => Velocity= 10.8m/s

Set slave ID

E.g., set slave ID=2, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD0	0x00 0x02	0x08	0x86

Sensor responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD0	0x00 0x02	0x08	0x86

Set baud rate

E.g., set baud rate to 9600, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD1	0x00 0x02	0x59	0x46

Sensor responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD1	0x00 0x02	0x59	0x46

Enquiry slave ID

Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0xFF	0x03	0x07	0xD0	0x00	0x01	0x91	0x59

Sensor responds:

Address	Function Code	Number of Points	address	Error Check (Lo)	Error Check (Hi)
0xFF	0x03	0x02	0x00 0x01	0x50	0x50